

SE 811: Software Maintenance

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Impact Analysis

Impact Analysis

- A change request (CR) activates an organization's process to modify a software system to carry out maintenance.
- The maintenance process is started by performing impact analysis.
- Impact analysis basically means identifying the components that are impacted by the CR.
- Impact analysis enables understanding and implementing changes in the system.
- Potential effects of the proposed changes are made visible by performing impact analysis.
- In addition, it is used in estimating cost and planning a schedule.

Impact Analysis

Before executing a change request, impact of the changes are analyzed for the following reasons:

- To estimate the cost of executing the change request.
- To determine whether some critical portions of the system are going to be impacted due to the requested change.
- To record the history of change-related information for future evaluation of changes.
- To understand how items of change are related to the structure of the software.
- To determine the portions of the software that need to be subjected to regression testing after a change is effected.

Impact Analysis

Impact analysis

- the assessment of the impact of a change to the source code of a module on the other modules of the system.
- the task of assessing the effects of making a set of changes to a software system.
- identifying potential consequences of a modification or discovering the entities to be modified to accomplish a change.

It determines the scope of a change and provides a measure of its complexity.

Impact Analysis Techniques

- Three classical impact analysis techniques, based on program dependency, are:
 - call-graph-based analysis,
 - static program slicing, and
 - dynamic program slicing.
- For example, inserting changes in the code and performing forward slicing from the changed points yield a set of statements that may be affected by the changes.

Impact Analysis Process

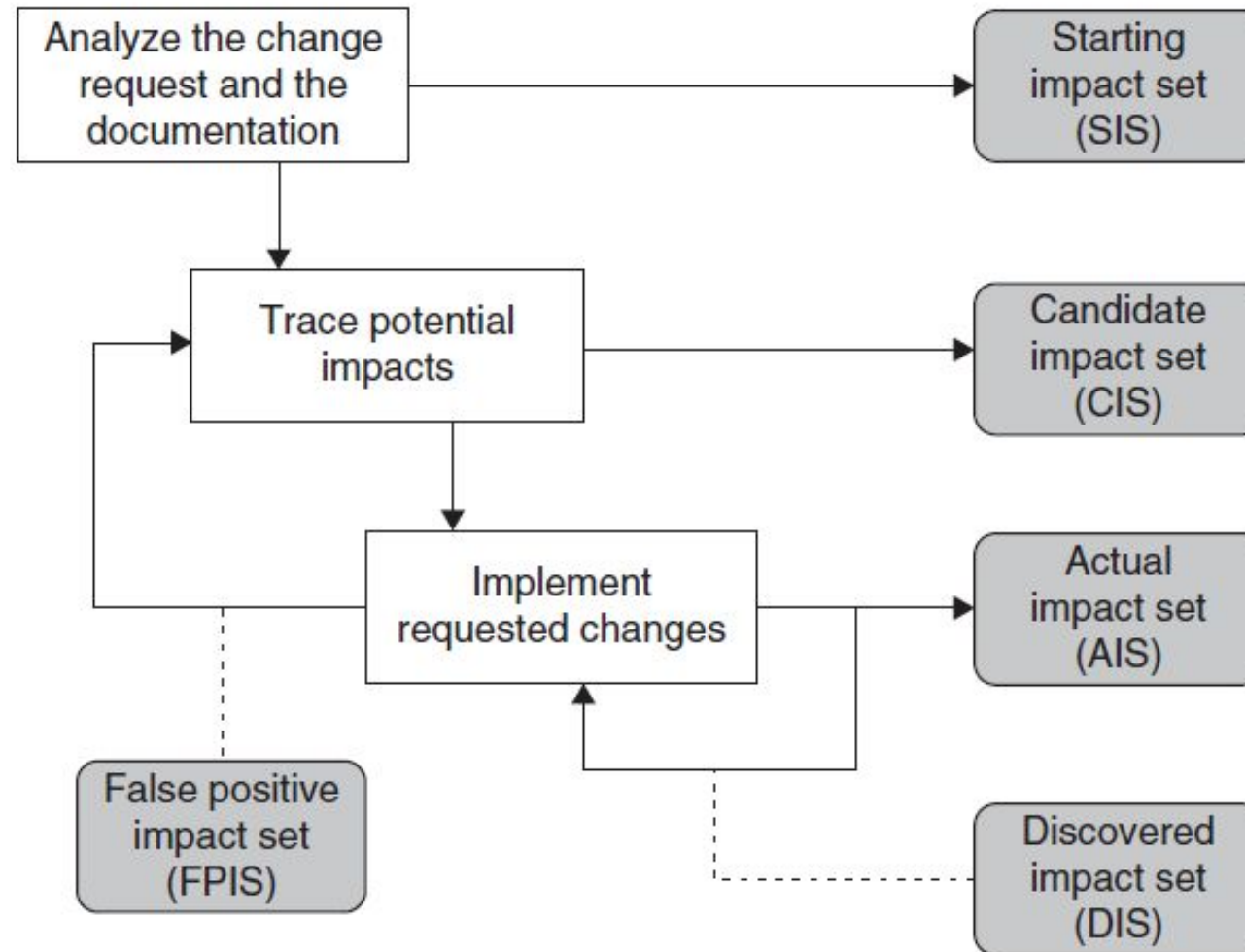


FIGURE 6.1 Impact analysis process. From Reference 6. © 2008 IEEE

Impact Analysis Terminologies

- *Starting Impact Set (SIS)*: The initial set of objects (or components) presumed to be impacted by a software CR is called SIS.
- *Candidate Impact Set (CIS)*: The set of objects (or components) estimated to be impacted according to a certain impact analysis approach is called CIS.
- *Discovered Impact Set (DIS)*: DIS is defined as the set of new objects (or components), not contained in CIS, discovered to be impacted while implementing a CR.

Impact Analysis Terminologies

- *Actual Impact Set (AIS)*: The set of objects (or components) actually changed as a result of performing a CR is denoted by AIS.
- *False Positive Impact Set (FPIS)*: FPIS is defined as the set of objects (or components) estimated to be impacted by an implementation of a CR but not actually impacted by the CR.
 - Precisely, $FPIS = (CIS \cup SIS) \setminus AIS$.
- Ideally, AIS should be equal to $SIS \cup DIS \setminus FPIS$,
- The error in impact estimation can be computed as $(|DIS| + |FPIS|)/|CIS|$.

Evaluating Impact Analysis Process

- *Recall*: It represents the fraction of actual impacts contained in CIS, and it is computed as the ratio of $|\text{CIS} \cap \text{AIS}|$ to $|\text{AIS}|$.
 - The value of recall is 1 when DIS is empty.
- *Precision*: It represents the fraction of candidate impacts that are actually impacted, and it is computed as the ratio of $|\text{CIS} \cap \text{AIS}|$ to $|\text{CIS}|$.
 - For an empty FPIS set, the value of precision is 1.

Identifying the SIS

- Concept assignment problem: map a new CR's “concepts” onto source code components (or objects).
- One discovers human-oriented concepts and assigns them to their realization.
- It is difficult to fully automate the concept assignment problem because programs and concepts do not occur at identical levels of abstractions, thereby necessitating human interactions.
- There are several methods to identify concepts, or features, in source code: grep tool, software reconnaissance, dependency-graph-based feature location, etc.

Software Reconnaissance

- The software reconnaissance methodology is based on the idea that some programming concepts are selectable, because their execution depends on a specific input sequence.
- Selectable program concepts are known as features. By executing a program twice, one can often find the source code implementing the features:
 - (i) execute the program once with a feature and once without the feature;
 - (ii) mark portions of the source code that were executed the first time but not the second time; and
 - (iii) the marked code are likely to be in or close to the code implementing the feature.

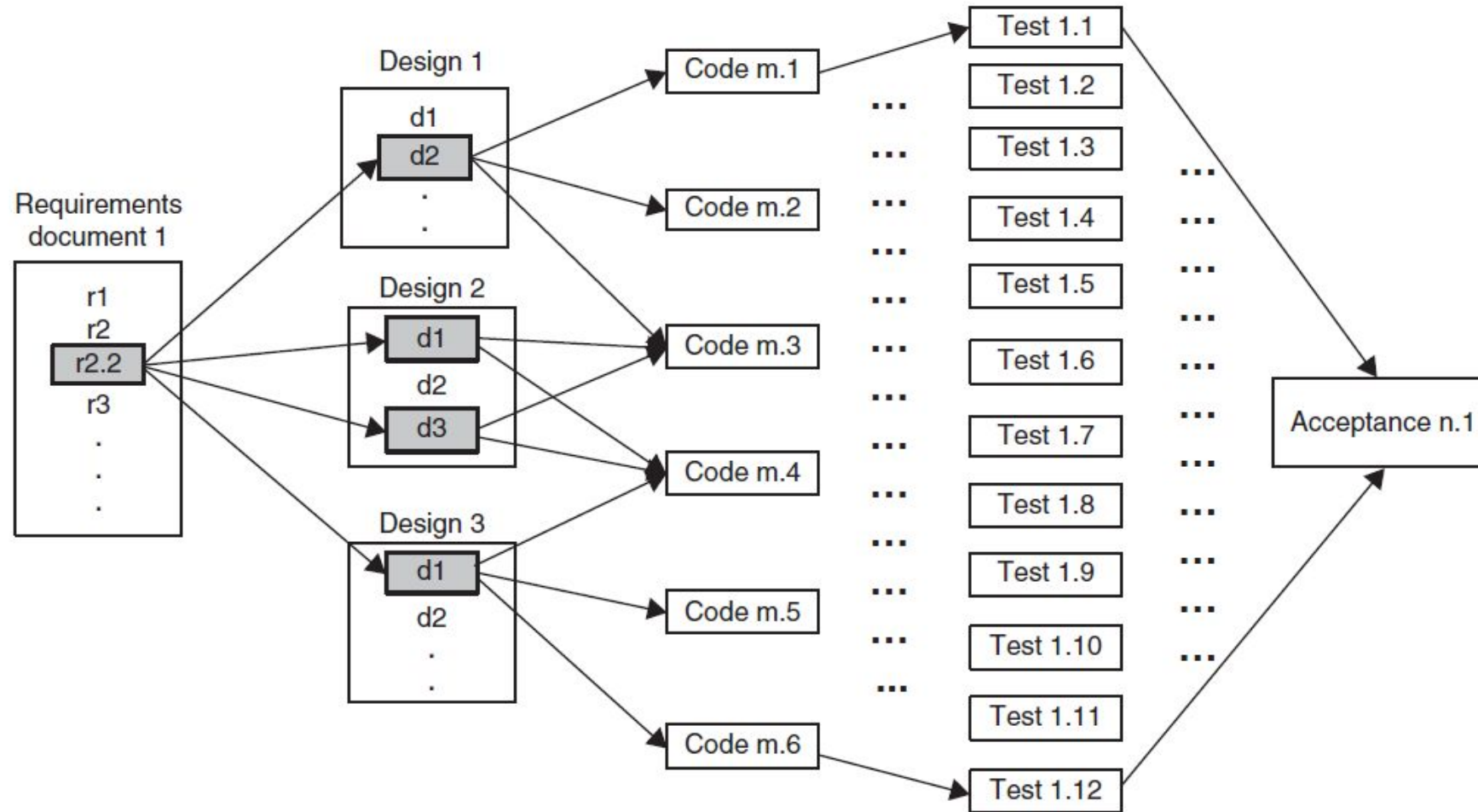
Artifacts Traceability

- Traceability is the ability to describe and follow the life of an artifact in both the forward and backward directions.
- It is the ability to trace between software artifacts generated and modified during the software product life cycle.
- It helps software developers understand the relationships among all the software artifacts in a project.
- After identifying the high-level documents about the feature to be modified, by using the concept of traceability, the maintainer locates the entities that need to be changed.

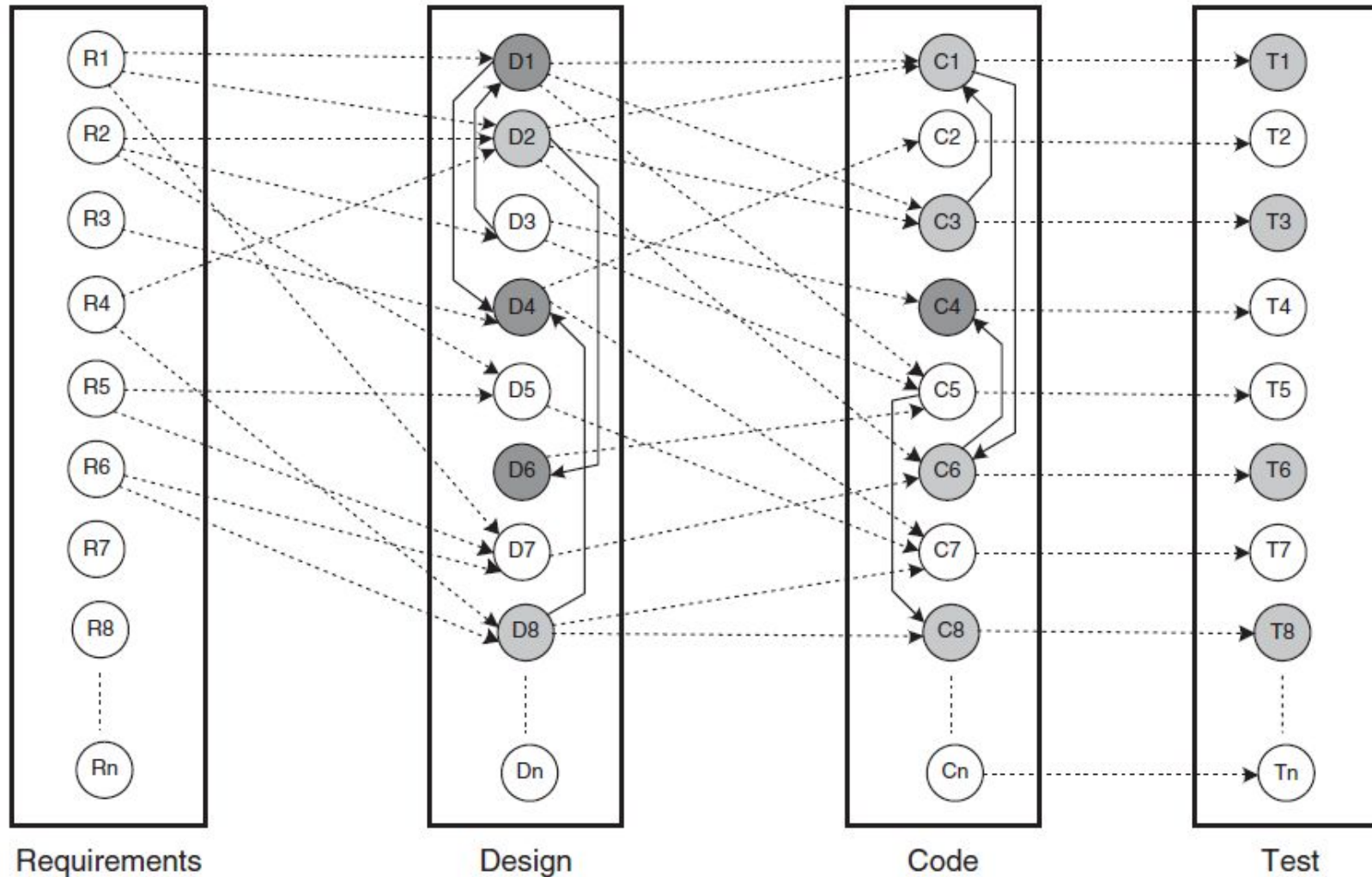
Artifacts Traceability

- There are two broad kinds of traceability:
 - (i) horizontal (external) traceability; and
 - (ii) vertical (internal) traceability.
- Traceability of artifacts between different models is known as external traceability, whereas internal traceability refers to tracing dependent artifacts within the same model.

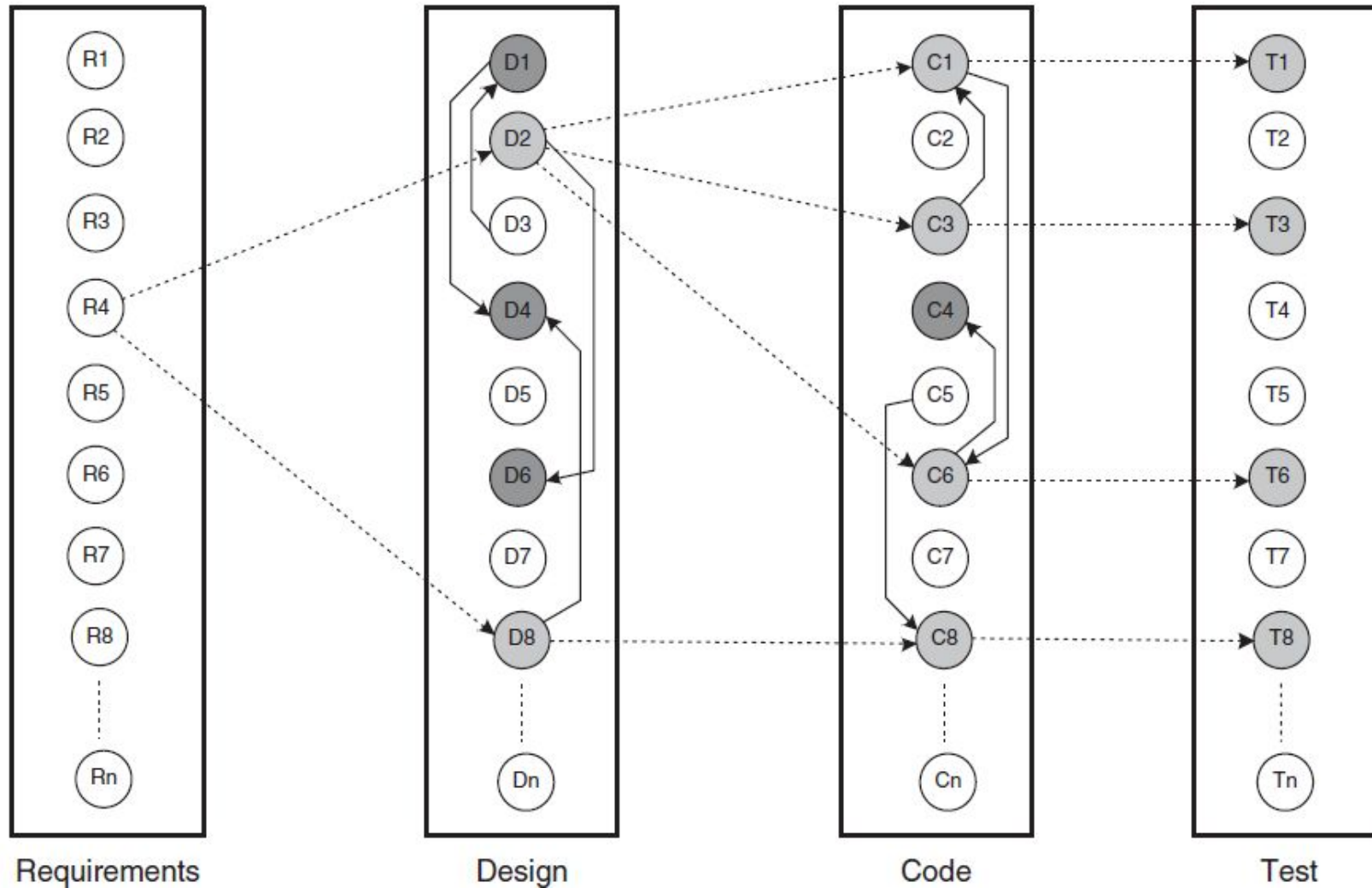
Analysis of Traceability Graph



Analysis of Traceability Graph



Analysis of Traceability Graph



Identifying the Candidate Impact Set

- The SIS is augmented with software lifecycle objects (SLOs) that are likely to change because of changes in the elements of the SIS.
- Changes in one part of the software system may have direct impacts or indirect impacts on other parts.
- ***Direct impact*** : A direct impact relation exists between two entities, if the two entities are related by a fan-in and/or fan-out relation.
- ***Indirect impact*** : If an entity *A* directly impacts another entity *B* and *B* directly impacts a third entity *C*, then we can say that *A* indirectly impacts *C*.

Identifying the Candidate Impact Set

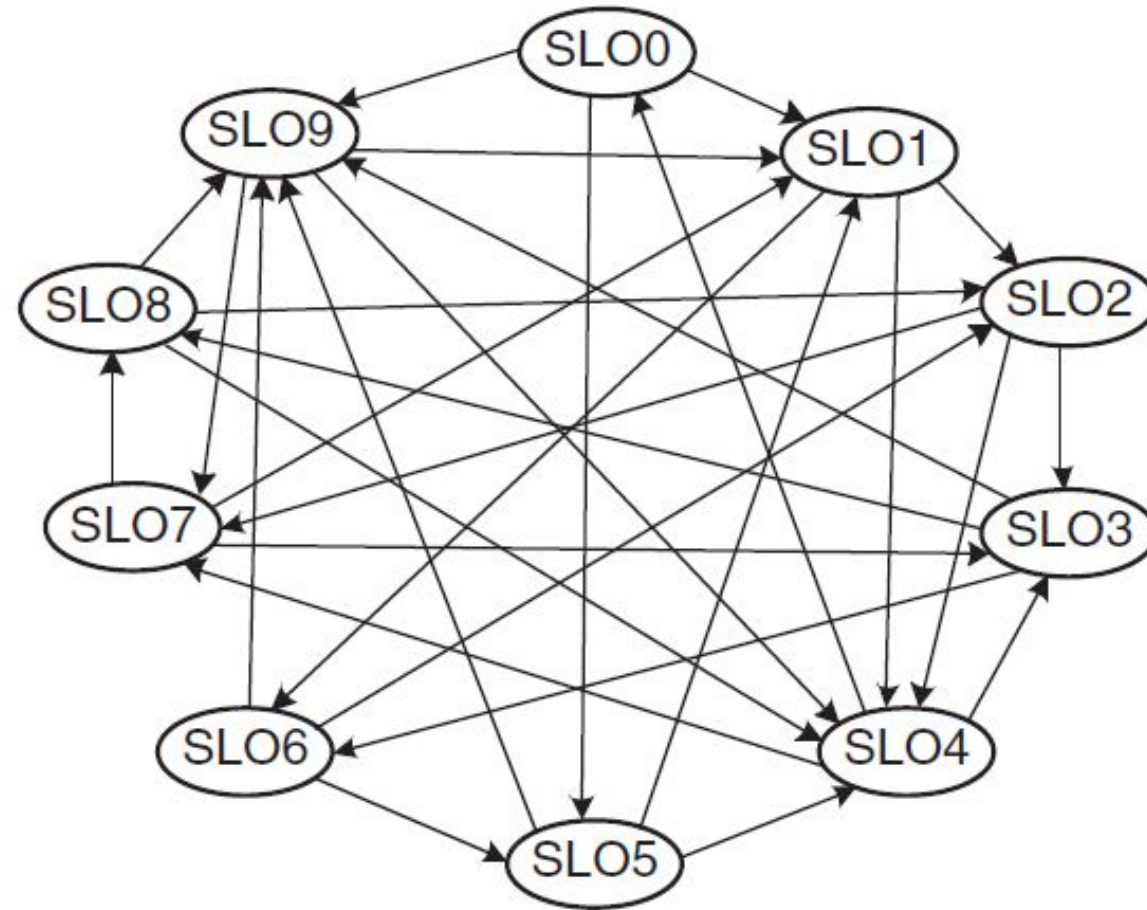


FIGURE 6.5 Simple directed graph of SLOs. From Reference 12. © 2002 IEEE

Identifying the Candidate Impact Set

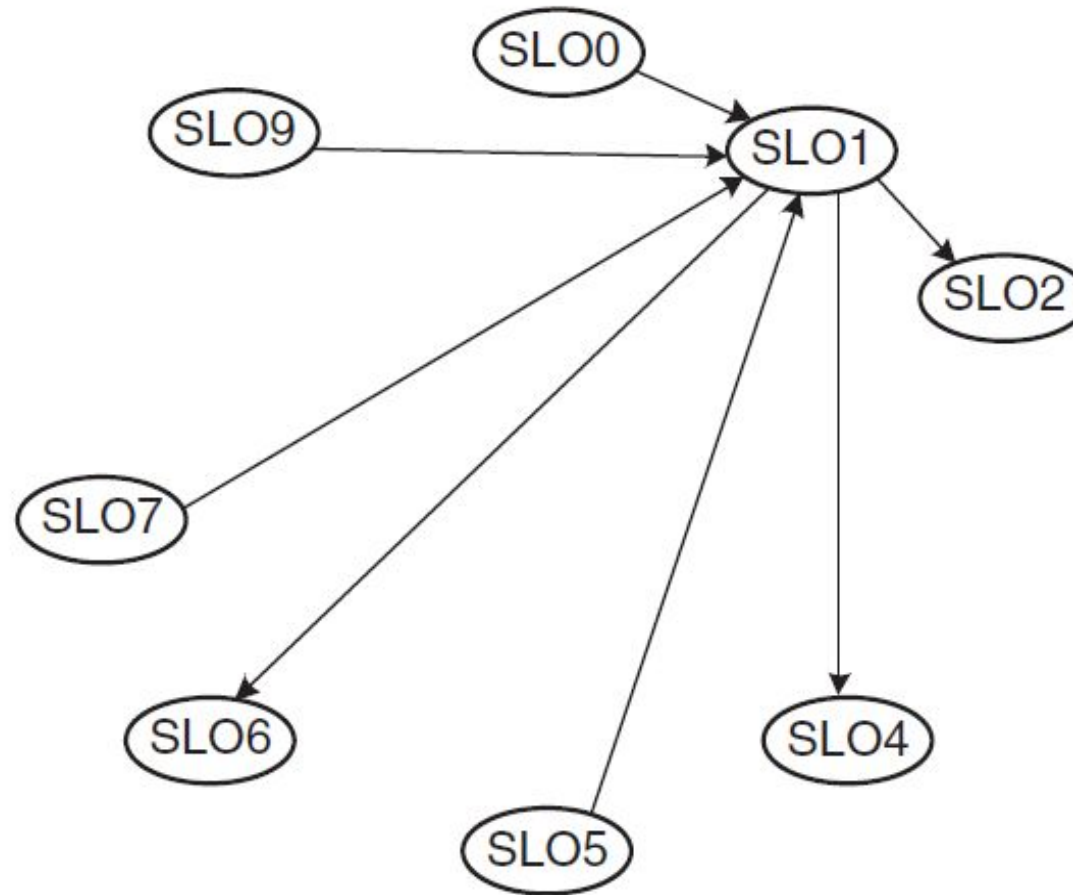


FIGURE 6.6 In-degree and out-degree of SLO1. From Reference 12. © 2002 IEEE

Identifying the Candidate Impact Set

TABLE 6.1 Relationships Represented by a Connectivity Matrix

	SLO0	SLO1	SLO2	SLO3	SLO4	SLO5	SLO6	SLO7	SLO8	SLO9
SLO0		X				X				X
SLO1			X		X		X			
SLO2				X	X			X		
SLO3							X		X	X
SLO4	X			X				X		
SLO5		X			X					X
SLO6			X			X				X
SLO7		X		X					X	
SLO8			X		X					X
SLO9		X			X			X		

Source: From Reference 12. © 2002 IEEE.

Identifying the Candidate Impact Set

TABLE 6.2 Relationships Represented by a Reachability Matrix

	SLO0	SLO1	SLO2	SLO3	SLO4	SLO5	SLO6	SLO7	SLO8	SLO9
SLO0		X	X	X	X	X	X	X	X	X
SLO1	X		X	X	X	X	X	X	X	X
SLO2	X	X		X	X	X	X	X	X	X
SLO3	X	X	X		X	X	X	X	X	X
SLO4	X	X	X	X		X	X	X	X	X
SLO5	X	X	X	X	X		X	X	X	X
SLO6	X	X	X	X	X	X		X	X	X
SLO7	X	X	X	X	X	X	X		X	X
SLO8	X	X	X	X	X	X	X	X		X
SLO9	X	X	X	X	X	X	X	X	X	

Source: From Reference 12. © 2002 IEEE.

Identifying the Candidate Impact Set

TABLE 6.3 Relationship with Distance Indicators

	SLO0	SLO1	SLO2	SLO3	SLO4	SLO5	SLO6	SLO7	SLO8	SLO9
SLO0		1	2	3	2	1	2	2	3	1
SLO1	2		1	2	1	2	1	2	3	2
SLO2	2	2		1	1	3	2	1	2	2
SLO3	3	2	2		2	2	1	2	1	1
SLO4	1	2	3	1		2	2	1	2	2
SLO5	2	1	2	2	1		2	2	3	1
SLO6	3	2	1	2	2	1		2	3	1
SLO7	3	1	2	1	2	3	2		1	2
SLO8	3	2	1	2	1	3	3	2		1
SLO9	2	1	2	2	1	3	2	1	2	

Source: From Reference 12. © 2002 IEEE.

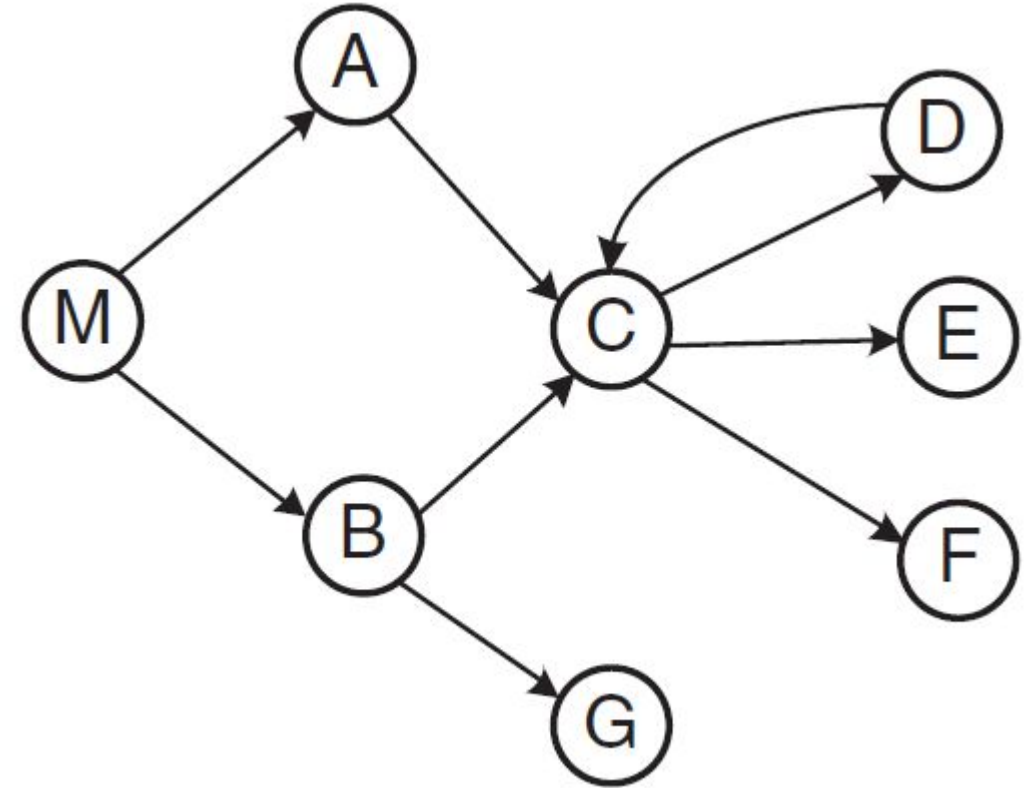
Identifying the Candidate Impact Set

- To minimize the occurrences of false positives, one might consider the following two approaches.
- **Distance-based approach:** In this approach, SLOs which are farther than a threshold distance are considered not to be impacted by changes.
- **Incremental approach:** In this approach, the CIS is incrementally constructed. For every SLO in the SIS, one considers all the SLOs interacting with it, and only SLOs that are actually impacted by the change request are put in the CIS. The identification process is recursively executed until all the impacted SLOs are identified.

Dependency-based Impact Analysis

Call Graph

- A *call graph* is a directed graph in which a node represents a function, a component, or a method, and an edge between two nodes *A* and *B* means that *A* may invoke *B*.
- Programmers use call graphs to understand the potential impacts that a software change may have



Execution Trace

M B r A C D r E r r r x

Dependency-based Impact Analysis

- **Program Dependency Graph**
 - static program slice
 - dynamic program slice